

CHAPTER 1

Coil Gun

PROJECT SIZE:	Medium
SKILL LEVEL:	★★★☆☆

THIS COIL GUN ([Figure 1-1](#)) will fire a small metal projectile at up to 30 miles per hour. It is portable, being powered by batteries, and is guaranteed to strike fear into the enemies of the Evil Genius. On dark evenings, the Evil Genius likes to strap flashlights to the heads of his minions and make them run around the Evil Genius' Lair while he takes potshots at them. Oh, how they squeal in panic!

A coil gun works in a similar way to a photographic flash gun. Capacitors are charged up over a few seconds, and then all their electrical energy is released extremely quickly. In a flash gun, the energy is released through a flash tube, and in a coil gun it is released into a coil of wire. This creates a powerful magnetic field that will cause any iron object near the coil to move.

Since the coil is wrapped around the tube from a plastic pen, and the iron projectile is inside the tube, it will fly along the tube towards the coil. As all the energy from the capacitors will be spent in a matter of milliseconds, the coil should ideally be turned off by the time the projectile passes its center and exits out the other side of the tube.

A similar, but less attractive analogy to the coil gun is a toilet cistern. In this case, the tank is like the capacitor, except that it is filled with water rather than charge. The tank charges over a period of a few tens of seconds. When the toilet is flushed, all the water rushes out.

The gun is controlled from a single three-position switch. When in its center position, the gun is off. When pushed forward, it starts to charge and the charging LED comes on. When fully charged, the LED goes off and the gun is ready for firing by pulling the switch back like a trigger.

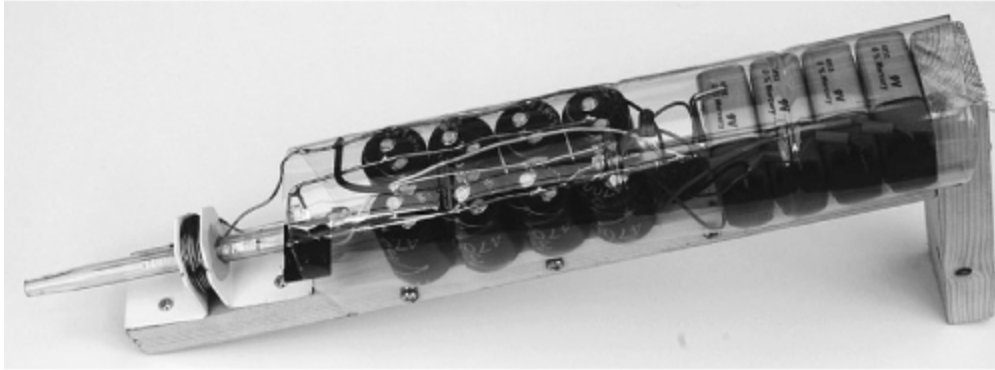


Figure 1-1 The coil gun

WARNING!

This is not a real gun. You could probably throw the projectile as fast as it comes out of the end of the gun. In addition, the projectile is lightweight.

However, this project has a number of dangerous aspects.

High currents. These are not high voltages, but the currents are very high and produce strong magnetic fields, so do not build this project or use it anywhere near anyone with a pace-maker.

Do not short-circuit the capacitors when they are charged up. You may melt whatever you are shorting them with, which means there will be small quantities of molten metal flying around.

And do not place your eye or anyone else's in the line of fire of this gun.

What You Will Need

The components for this coil gun are all readily available. It's worth shopping around for the capacitors; eBay usually has a good selection of suitable ones if you search for "electrolytic capacitors." You'll need the parts in the Parts Bin.

You will also need the following tools listed in the Toolbox.

TOOL BOX

- Soldering equipment
 - Hacksaw
 - Wood saw
 - Drill and assorted drill bits
 - Epoxy resin glue or hot glue gun
 - Multimeter
-

PARTS BIN			
Part	Quantity	Description	Source
Firing tube	1	Disposable transparent ballpoint pen about $\frac{3}{4}$ of an inch (7mm) in diameter	
Coil retainers	2	Plastic brackets cut from a plastic food container (see the following description)	
Projectiles	2	Iron/steel nails $\frac{1}{8}$ inch (3mm)	Hardware store
Lumber		18 inches (45cm) length of $\frac{3}{8}$ " $\times$ 1 $\frac{1}{4}$ " (18mm $\times$ 33mm) wood	Hardware store
Plastic drink bottles	1		
Plastic insulating tape			Hardware store
Coil wire		13 feet (4m) of 20 or 21 AWG enameled copper wire	Farnell: 1230984
Batteries	4	Budget PP3 9V batteries	
Battery clips	4	PP3 battery clips	Farnell: 1183124
C1-8	8	4700 μ F 35V electrolytic capacitors, or any set of 35V capacitors totaling around 38,000 μ F	Farnell: 9452842 eBay
SCR	1	40TPS12A Thyristor 55A	Farnell: 9104755
S1	1	DPDT On-off-momentary toggle switch	Farnell: 9473580
R1	1	100 Ω 2W	Farnell: 1129029
R2	1	100 Ω 0.5W	Farnell: 1127903
D1	1	5mm Red LED	Farnell: 1712786
R3	1	2.7K Ω 0.5W	Farnell: 9338667
D2	1	5.1V 5W Zener diode	Farnell: 1705663

Assembly

The schematic diagram for the project is shown in [Figure 1-2](#). Only a few components can be soldered together without the need of a circuit board.

The design has three main sections: the capacitor bank, the trigger circuit, and the charge indicator LED. The capacitors are all connected in parallel and charged by the batteries through the 100 Ω resistor when the switch is in the “charge” position.

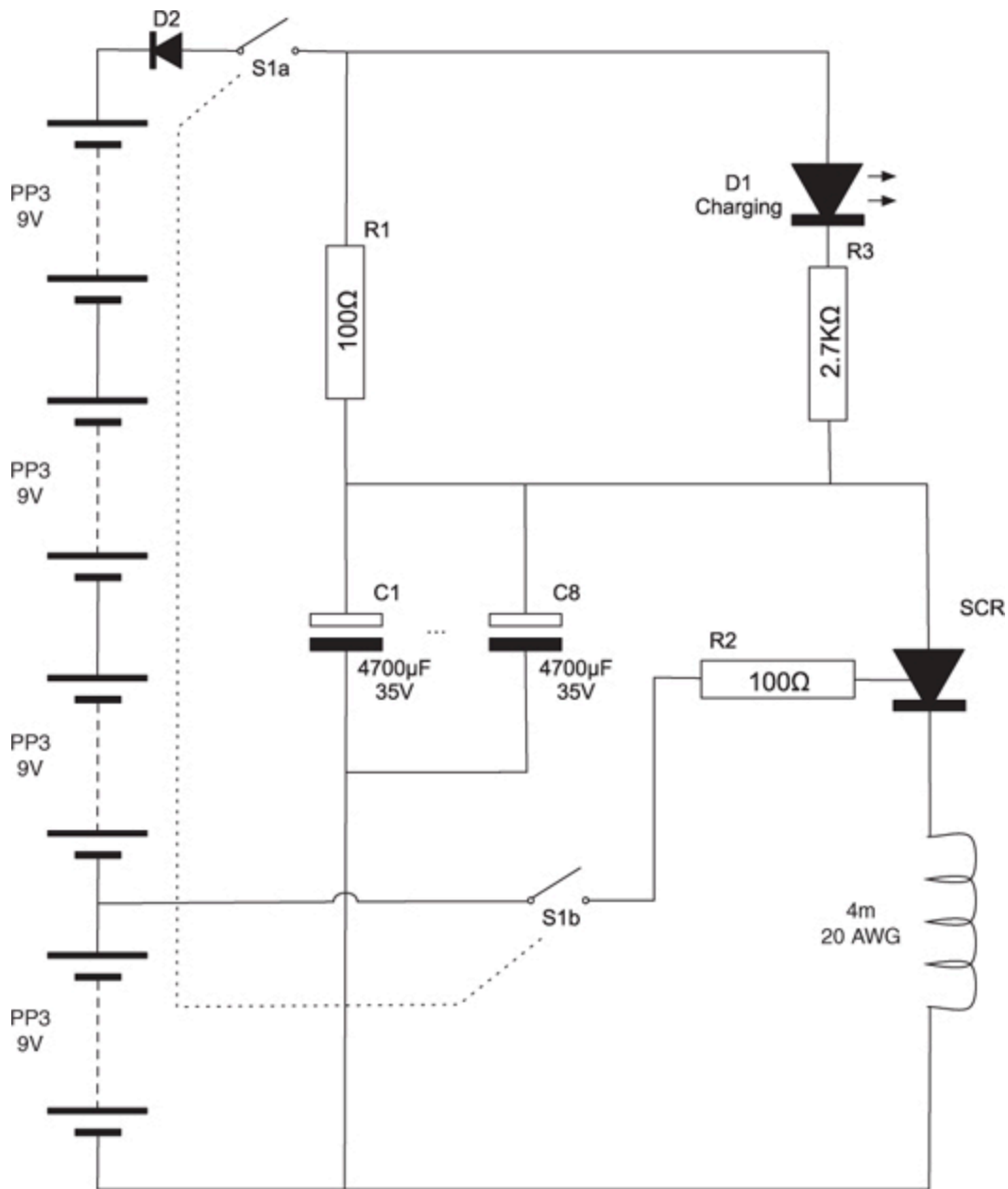


Figure 1-2 The schematic diagram

The trigger circuit uses a SCR (silicon-controlled rectifier), or Thyristor as they are sometimes called. The SCR acts as a conducting switch when a current passes through its “gate” connection. This happens when the toggle switch is put to its “fire” position.

Figure 1-3 shows the full wiring diagram for the coil gun. To maintain a “gun-like” shape, the components are laid out in a line, with the batteries at the back and the coil at the front.

Step 1. Make the Coil Former

The coil is wound onto a disposable ballpoint pen tube. The narrow end of the tube, where the nib would go, should be sawn off and the bung removed from the other end of the tube. To hold the windings of the coil in place, we use rightangle plastic brackets cut from a food container (**Figure 1-4**). Any kind of plastic with a 90-degree bend can be used here.

The brackets should be about 1 inch (25mm) on each side. They are then drilled in the center so that they fit snugly over the tube. Drill a small hole on one of the brackets, immediately adjacent to the large hole (**Figure 1-5**). This is where the innermost connection to the coil will emerge, so the hole must be just big enough for the coil wire. The brackets should then be fixed in place, leaving a gap of about $\frac{3}{8}$ of an inch (10mm) between them (see **Figure 1-6**). Epoxy resin glue or glue from a hot glue gun is used to hold the discs in position.

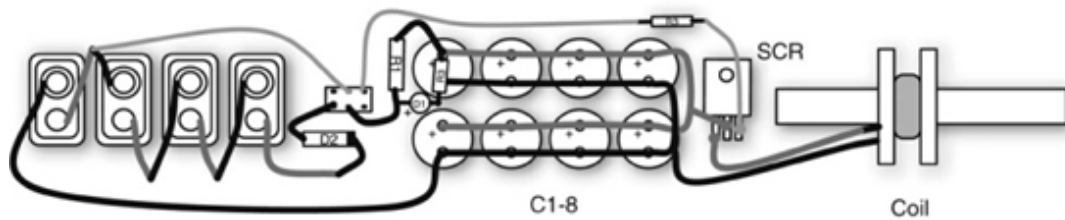


Figure 1-3 The wiring diagram

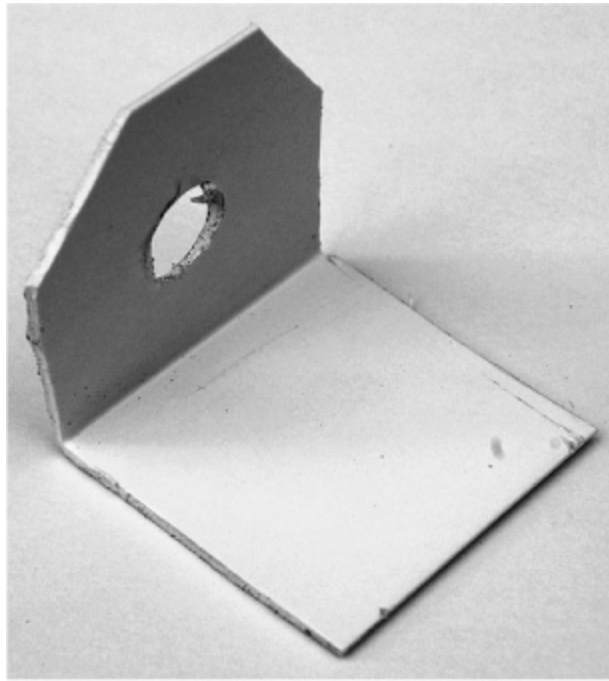


Figure 1-4 Bracket to hold the firing tube

Step 2. Wind the Coil

The coil is made up of 13 feet (4m) of 20 AWG enameled copper transformer wire. Winding coils by hand is a little tedious. Fortunately, this coil uses a short length of wire. It is worth trying to wind the coil neatly, but it usually goes off the rails as you get toward the end. This does not really matter.

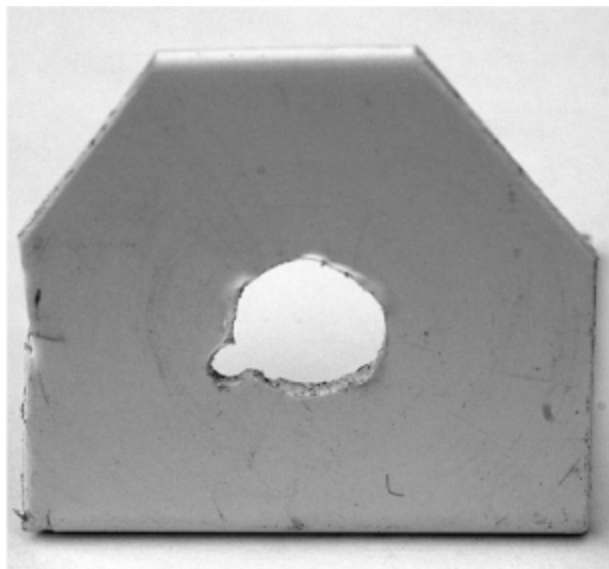


Figure 1-5 Drilling the bracket

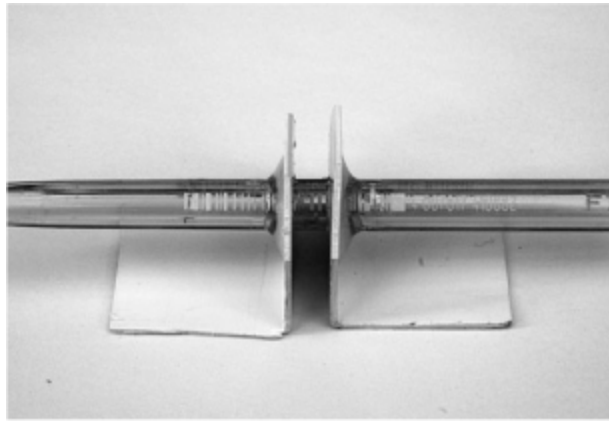


Figure 1-6 The coil former

Start by measuring out 13 feet (4m) of wire. Thread about 2 inches (50mm) of the wire through the small hole made in the disc, close to the pen tube. This will be one lead of the coil. Then coil the wire around the pen, keeping each turn as close as possible to the previous turn ([Figure 1-7](#)).

When you get to the disc at the other end, keep winding in the same direction but allow the turns to line up back toward where you first started. Try to keep the turns as close together and as tight as possible. It can help to, from time to time, put a drop of superglue onto the coil to hold the turns into place.

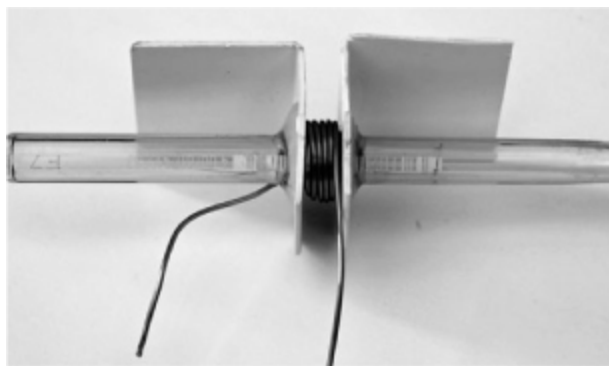


Figure 1-7 Winding the coil

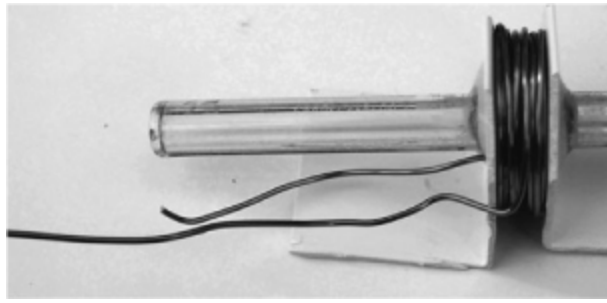


Figure 1-8 The finished coil assembly

You should end up with about seven layers on the coil (**Figure 1-8**). Leave about 2 inches (50mm) free and carefully cut a slot in the edge of the bracket for the free end of the coil. Then, add a bit more glue to make sure the coil stays together.

Later on, we will need to solder the ends of the coil, so scrape the enamel off the ends of the wire and coat the ends with solder.

Step 3. Assemble the Capacitor Bank

The capacitors used in this project were selected to provide the most farads per buck. We used $8 \times 4700\mu\text{F}$ (microfarad) capacitors rated at 35V. This gave us a total of $37,600\mu\text{F}$. Four $10,000\mu\text{F}$ capacitors will work just as well, if not slightly better, when it comes to holding a little more charge. However, you must make sure that the voltage rating is 35V or more.

You should also avoid the temptation to greatly increase the capacitance, as this will increase the maximum current, which may be too much for the SCR. You may wish to experiment with this, but do understand that when the current becomes too much, it will destroy the SCR.

Figure 1-9 shows how the capacitors are connected together into two rows. It is easiest to make each row of four first and then connect the two rows together.

Start by lining up four capacitors on their backs with their legs in the air. Make sure that all the negative leads are on one side and all the positive on the other. It's very important that the capacitors are connected the right way around. If one of the capacitors is the wrong way around, it could explode—and capacitors are full of messy goo!

Now take some solid core wire and connect all the negative leads together, and then do the same for the positive leads. You can use the same wire as you used to wind the coil, but you will need to scrape away the insulating enamel where you want to make a solder joint. I used solid core wire of the type employed in domestic electrical wiring. This has the added advantage of being able to use the plastic insulation to color-code the positive and negative connections to the capacitor bank.

Use the thickest wire you can get your hands on. This wire is going to carry a current of around 100A and the thicker the wire, the lower the resistance and the more energy will be transferred into the coil.

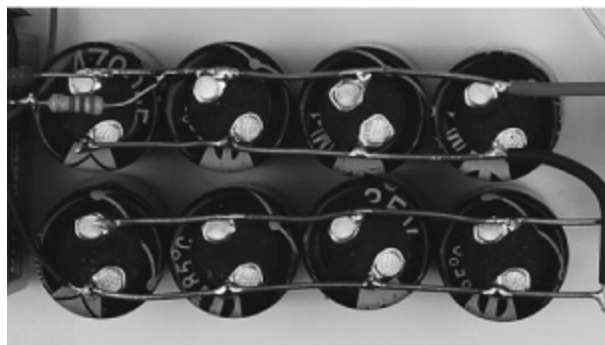


Figure 1-9 Constructing the capacitor bank

When both rows of four capacitors are complete, you need to join the common positive connection of one bank to the common positive of the other bank. Do the same for the negative connections (refer back to the wiring diagram of [Figure 1-3](#)).

Step 4. Add the Triggering SCR

You may be wondering why we need to use a SCR and why we couldn't just use the switch directly between the capacitors and the coil. The answer is that no regular switch would withstand the hundreds of amps that flow when the coil is triggered. It would simply weld the contacts together or melt them.

The SCR that we have chosen is a good compromise between power handling and price. It will allow peak currents of up to 500A for a millisecond. We are going to need it to handle about 100A for 10 milliseconds.

The SCR sits between the capacitors and the coil ([Figure 1-10](#)). The 100 Ω resistor is connected to the gate connection.

The middle connection to the SCR is connected to the positive side of the capacitor bank, and the leftmost connection to one side of the coil. The other side of the coil is connected to the negative side of the capacitor bank (see [Figure 1-10](#)).



Figure 1-10 The SCR and gate resistor

Step 5. Fitting the Batteries and Switch

The four 9V batteries are connected in series to give a total of 36V. However, fresh batteries may have upwards of 10V per battery (a total of 40V), which is above the rated voltage of the capacitors. To play it safe, the Zener diode in series reduces this by about 5.6V, bringing the

voltage just below the capacitors' rated voltage. Note that exceeding the rated voltage of electrolytic capacitors is dangerous and will shorten the life of the capacitors.

Cut the battery leads so they are a more manageable length and then connect the positive (red) lead of the first battery lead to the negative lead (black) of the second lead, and so on. Finally, connect the Zener diode in series between the last positive lead and one side of the switch, as shown in **Figure 1-11**.

The switch is what is called a double-throw switch (**Figure 1-12**). That is, it is actually two switches operated by a single lever. One of the “throws” of the switch is used to turn charging on and off, and the other is used as a trigger. We now need to connect the “trigger” throw of the switch to R3, which we have already soldered to the gate of the SCR (see **Figure 1-3**). The other connection to the trigger throw of the switch should be connected to the positive connection of the first battery. For convenience, that can be where the battery leads of the first and second batteries are soldered together.



Figure 1-11 The batteries and switch

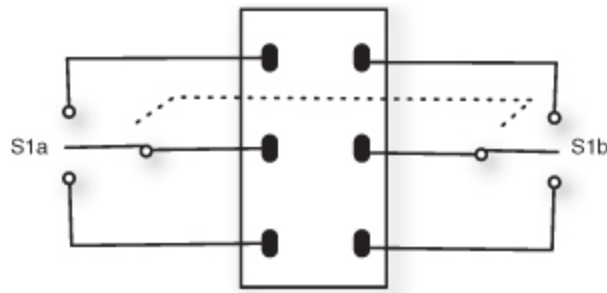


Figure 1-12 Double-throw switch

The switch used has three positions. The center position turns both halves of the switch off. This is the position shown in [Figure 1-12](#). Push the toggle one way and the switch will latch on. This will be the charging position. Pulling the toggle lever the other way will connect the switch momentarily, but the switch is sprung to pull the switch back to the center off position.

So, when connecting the switch you need to make sure it is the right way around so that the firing circuit is switched when it goes into its non-latching action and the charging circuit is made when the switch is in its latched mode. When the switch toggle is in the up position, this usually connects the center connection to the bottom pin, and vice versa. If you get it wrong the first time, you will find you have to hold the toggle in the momentary position to start charging. If this is the case, just unsolder all the leads to the switch and flip it through 180 degrees, then solder the leads up again in the same positions as they were before.

Step 6. The Charging LED

The charging LED will light when the capacitors are still filling with charge. Once they are full, it will go off and the gun will be ready for firing. It is just an LED with a series resistor to limit the current. It will start bright and gradually get dimmer as the capacitors get fuller.

[Figure 1-13](#) shows the LED and resistor soldered across the charging resistor. Note that the positive (slightly longer) lead of the LED must be

connected to the battery end of the charging resistor, and the negative end must be connected to the LED's current limiting resistor.

We have now soldered everything together, but before we start fitting things into a case, we need to test out our coil gun on the bench.

Step 7. Making the Projectiles

We need something for our coil gun to fire. Iron nails $\frac{1}{8}$ of an inch (3mm) in diameter are good for this, but they are a little long. Our projectiles should be about $\frac{1}{4}$ -inch (5mm) long (**Figure 1-14**). Since, these can be hard to find once fired, it is a good idea to have a few.

The Evil Genius has discovered that the best way to find lost projectiles is to take away the shoes and socks of his minions. While walking about barefoot, the fragments of nail invariably attach themselves to their feet.

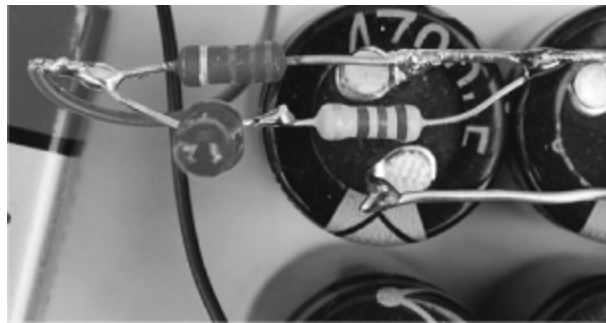


Figure 1-13 The charging LED and resistor

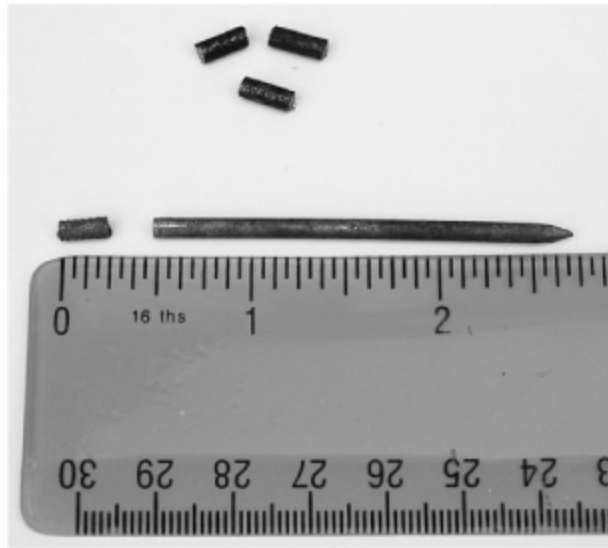


Figure 1-14 The projectiles

Use a hacksaw to cut the nail into pieces the right length.

Test Firing

Now we get to the exciting bit! Before we start, we need to check that everything is as it should be. We are using very high currents here, so there is the potential to destroy our components if we are not careful.

Basic Checks

Before connecting the batteries, compare all the wiring with [Figure 1-3](#) and make sure we have all our connections right. Once you are sure everything is OK, put the switch into its center off position and connect up the batteries.

We are going to start with a low voltage test before we ramp up to full power. Put your multimeter onto its 20-volt range, or at least enough to display 10V, and then connect the multimeter leads across the capacitor bank, as shown in [Figure 1-15](#).

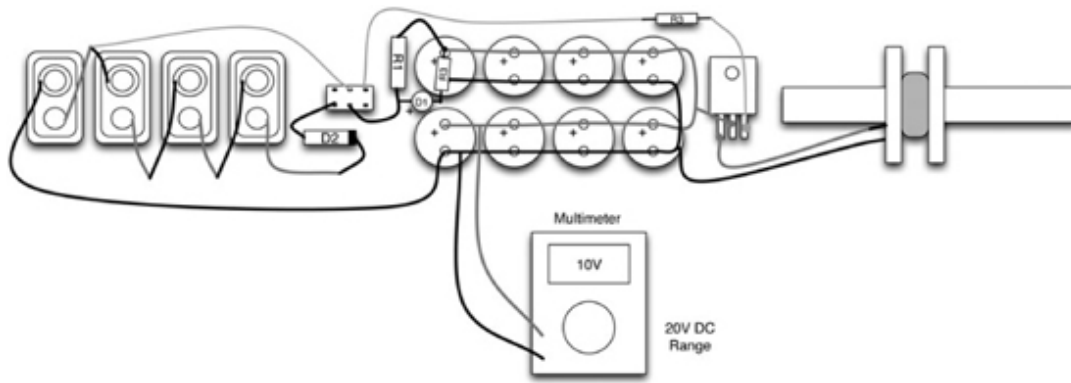


Figure 1-15 Measuring the charge voltage

Put the switch to the “charge” position. If the LED does not light, put it immediately back to the center off position and check your wiring. Also, check that your switch is wired the correct way around. You can test this last point by just putting the switch to the momentary “fire” position. If the LED lights, then your switch is probably the wrong way around (see the earlier section titled “Step 5. Fitting the Batteries and Switch”).

Note that, during charging, R1 will get hot.

As soon as the multimeter indicates about 10V, put the switch back to the center position. Watching the multimeter, you should see the voltage decrease very slowly. If it drops down to 0V quickly, then something is wrong, so put the switch back to off and check everything.

Assuming this test passed, we can now push the projectile into the firing tube until the end is just level with the edge of the coil on the far side of the coil. Now put the switch to fire and the projectile should move, hopefully traveling through the tube and emerging at a modest speed.

Congratulations! Since everything seems in order, it’s time to try a full-power test. This time, you can just let the gun charge until the LED turns off, which should be after 10 or 15 seconds.

Measuring the Projectile Speed

You can tell if the firing was a good one, and the projectile was going fast, just by observation, but it's better to have a more precise way of measuring the speed. The way to do this is with a humble computer. I am indebted to the excellent Barry Hansen for his web site at www.coilgun.info that, amongst many other useful things, describes the decidedly Genius (I would not say Evil) way of measuring the speed with nothing more than a computer with a microphone input.

Here is how it works. You just record the sound of your test and then use some sound software to examine the waveform and measure the time between the coil firing and the projectile hitting its target. I think it's safe to say that our projectiles are slow enough that we can ignore inaccuracies due to the speed of sound.

First of all, you will need to download some sound recording software. I used Audacity (<http://audacity.sourceforge.net>) because it is free and available for most operating systems, including Windows, LINUX, and Mac.

You then need to set up your coil gun a convenient distance away from a target that will make a noise when the projectile hits it—say, about 3 feet (1m). I use a plastic shopping bag hung from a door handle that is conveniently the same height as my workbench. The bag has the added advantage of absorbing most of the energy from the projectile, thus preventing it from bouncing off and becoming lost. This arrangement is shown in **Figure 1-16**.

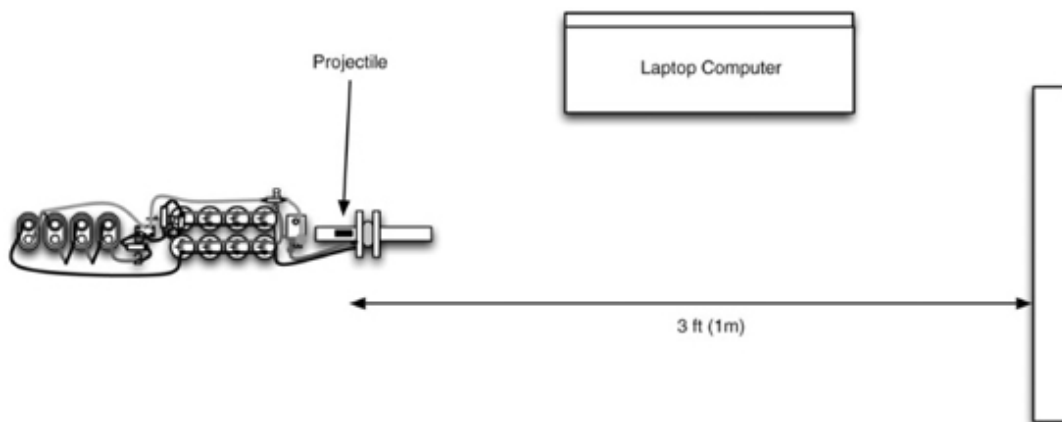


Figure 1-16 The test firing arrangement

Once your capacitors are fully charged, the LED turns off, and the voltage across the capacitors is about 35V, place the projectile just to the far side of the coil from the target, start your sound software recording and fire the gun. As soon as the gun has fired, stop the recording and look at the sound waveform.

After a bit of cropping and zooming on the resulting sound waveform, you should see something like **Figure 1-17**.

The human ear is a wonderful thing, and you can use yours to check that the sound spikes correspond to the firing and target impact by selecting one of the areas and clicking play to hear what it sounds like. From **Figure 1-17**, you can see that the time from the end of the sound of the triggering to the first sound of the impact is $0.15 - 0.085$, or 0.065 seconds.

You now need to measure the distance from the starting position of the projectile to the front of the target. We can calculate the velocity as the distance divided by the time. In this case, the distance was 1m and the time 0.065s. So the velocity was 15.39 meters per second. Multiply this by 2.237 to get a figure of 34.4 miles per hour.

I found it best to make a spreadsheet for the results, as shown in **Figure 1-18**. Successive test results can each be recorded on a separate row.

We now need to find the optimal starting position for the projectile. Using a thin pen that will write on the firing tube, draw a series of dots a few millimeters apart along the tube from the far side of the coil from which you want the projectile to emerge. Then, using the same projectile each time, measure the velocity at each of the positions. You can either line up the front or the back of the projectile with the dot. It does not matter as long as you are consistent.

Putting the Project in a Case

At the moment, our gun is not very portable, so we need to build a case for it. The components look quite impressive, so we are going to make a transparent case, but mount it onto a strip of wood ([Figure 1-1](#)). The plan for the wood is shown in [Figure 1-19](#). There are just two bits of wood, and the exact sizes are not critical. The main piece needs to be long enough to accommodate all the components laid out in a line, and the end piece serves the dual purpose of providing a handle and keeping the batteries in place. The end piece is drilled and screwed to the main piece.

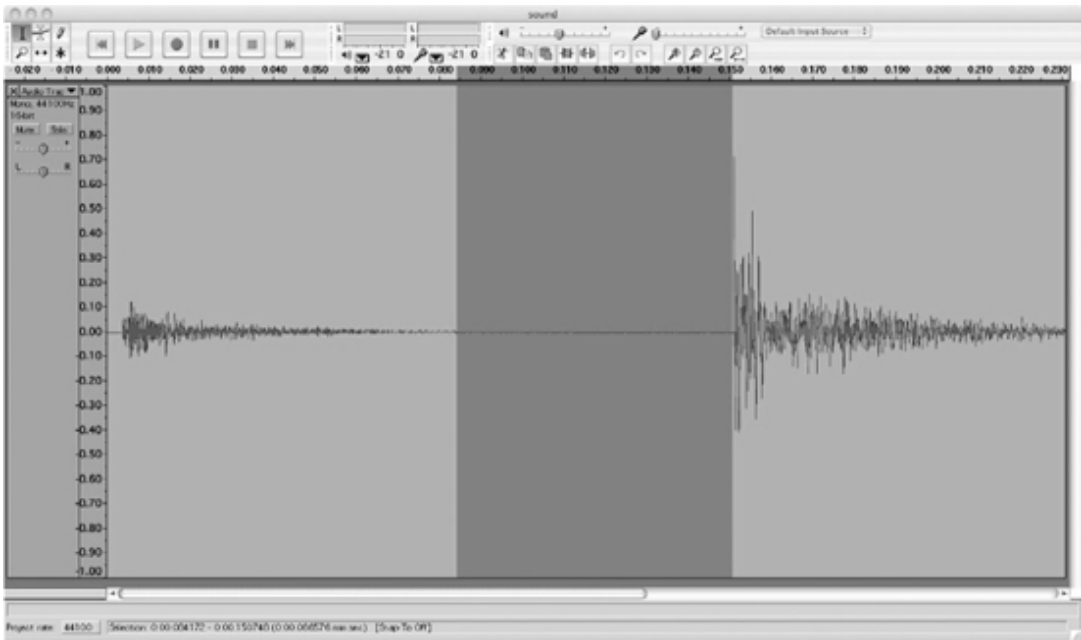


Figure 1-17 The sound waveform from a test firing

	A	B	C	D	E
1					
2					
3					
4	T1 (fire)	T2 (hit target) d (m)		v (m/s)	v (mph)
5	0.085	0.150	1.000	15.385	34.415
6					
7					
8					

Figure 1-18 A spreadsheet to calculate the velocity

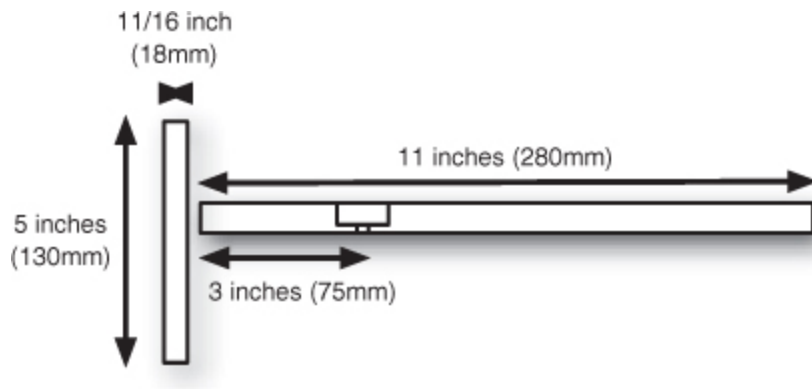


Figure 1-19 The wooden frame for the coil gun

Two holes should be drilled into each of the brackets of the firing-coil assembly so they can be screwed into the wood.

Figure 1-20 shows a close-up of the hole for mounting the switch. Drill most of the way through the wood—leave about $\frac{3}{16}$ of an inch (5mm) with a $\frac{1}{16}$ -inch (30mm) bit, which should make a hole large enough to accommodate the whole switch. Then, drill the remainder of the hole through to allow the neck and toggle of the switch to push all the way through. Afterward, fasten the switch on with its retaining nut.

This requires a bit of care to make sure you do not drill the bigger hole all the way through the wood.

Figure 1-21 shows the switch fitted like a trigger below the electronics, as well as the arrangement of the capacitors and other components on top of the wooden structure.

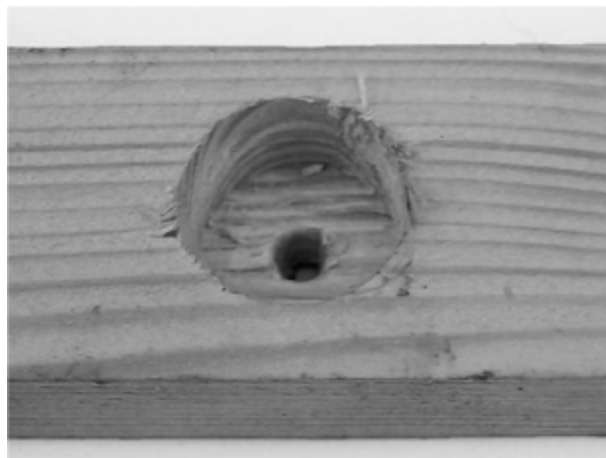


Figure 1-20 The hole for mounting the switch

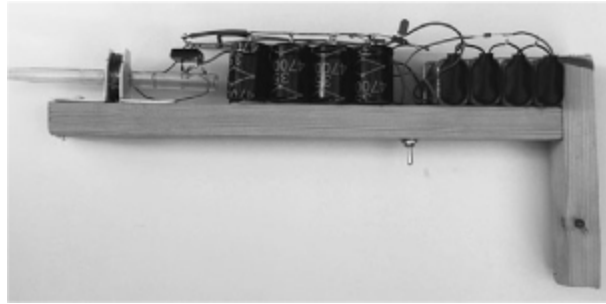


Figure 1-21 The switch mounted as a trigger

The top view of the whole project is shown in [Figure 1-22](#). Note how we have used a little brass hook to keep the batteries in place. You should also use self-adhesive pads or dabs of glue from a hot glue gun to keep the capacitors in position.

Before fitting the plastic cover, put some insulating tape over any bare wires that might move and touch something they shouldn't.

The final refinement—done to make the gun easier to use—is to make a “stop” to prevent the projectile from falling out of the back of the coil. This way we can just drop the projectile in from the front and know that it is in the correct position.

To do this, use a $\frac{1}{32}$ -inch (1mm) drill bit to make a tiny hole in the firing tube at the position where the end of the projectile furthest from the coil should be placed for best firing. You will have determined this from your earlier experiments. Then, put a short length of $\frac{3}{8}$ -inch (about 1mm) wire (I used a bit of resistor lead) through the hole and bend over both ends so it stays in place. You can see this in [Figure 1-23](#).

An optional refinement to the case design is to cut some thin Perspex or other flexible plastic (say, from a large plastic bottle) and bend it over the wood from one side to the other, fixing it in place with screws.

Simply take a plastic drink bottle and cut off both ends, then measure out the right length and width of the curved bottle plastic to fit round the top of the gun. Refer back to [Figure 1-1](#) to see how this looks.

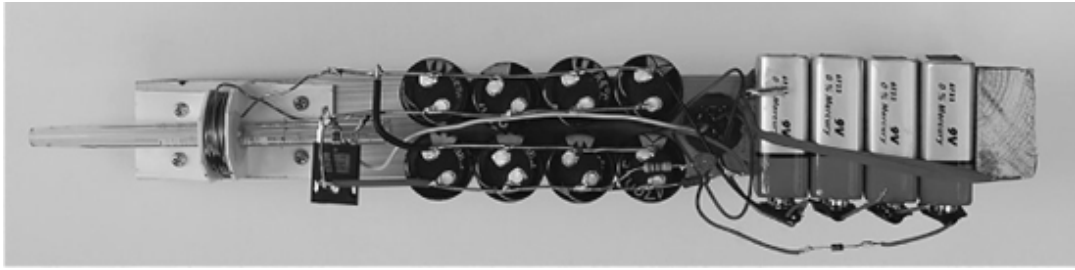


Figure 1-22 Top view showing everything in place

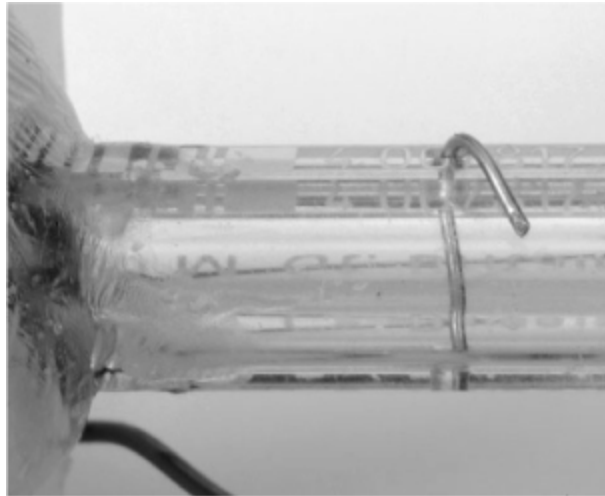


Figure 1-23 Close-up of the end stop

Theory

Coil guns are remarkably inefficient. You are doing well if 1 or 2 percent of the energy in the capacitors is converted into kinetic energy of the projectile. If you can weigh your projectile, or perhaps weigh a few of the nails it is made from, and then do the math, you can calculate the efficiency.

The energy stored in the capacitor is calculated by the formula:

$$E = (CV^2)/2$$

Where E is the energy in joules, C is the capacitance in farads, and V is the voltage.

So, for our arrangement of capacitors, the energy available for each firing is:

$$(0.0376 \times 352) / 2 = 23 \text{ joules}$$

This is similar in energy to the kinetic energy in a .177 air gun. So if we had 100 percent efficiency, our coil gun would be really quite dangerous.

So, now that we know that the “input” energy is 23 joules, let’s see how many joules of kinetic energy there are in the projectile.

To do this, we can use the formula:

$$E = (mv^2)/2$$

Where m is the mass of the projectile in kilograms, and v is the velocity in meters per second.

My projectiles weigh approximately 0.3 g and the best velocity I got was 15m/s. This velocity is actually the average velocity of the projectile’s flight to the test target, which will be a little less than the muzzle velocity, but it is probably close enough not to matter much.

So, the energy transferred from the capacitors to the projectile is:

$$(0.0003 \times 15 \times 15)/2, \text{ or } 0.033$$

This means that the efficiency of our coil gun is:

$$0.033/23, \text{ or } 0.14 \text{ percent}$$

Not brilliant. So where does all the energy go?

Some of the energy is lost because the projectile is first pulled into the coil, and then if the magnetic field has not disappeared before it passes the center of the coil, it will be pulled back into the coil, slowing it down. The answer to this is to shorten the pulse, which can be done in two ways:

- Use fewer capacitors (but this will reduce the input energy).
- Use less turns of wire for the coil (but this will increase the maximum current, potentially destroying the SCR).

The geometry of the coil and the type and size of the projectile, as well as the material it is made from, all affect performance. Getting the most out of your coil gun is very much a matter of trial and error and reading about other people's trial and error.

The Internet offers some great resources on this. The Wikipedia entry for coil guns is a great starting point, as is www.coilgun.info.

If you have an oscilloscope, you can use it to measure the duration of the pulse through the coil. You will need to set your scope as follows:

- **Channel sensitivity:** 5V/div. Make sure the scope is okay with 35V.
- **Trigger mode:** single shot, rising edge.
- **Timebase:** 2.5 ms/div.

You can see a firing of the gun in [Figure 1-24](#). This shows that at 2.5 ms/division, the pulse is about 15 ms in duration.

Summary

This is a fun project to make. As we discovered in the theory section, the gun is very inefficient. This is probably just as well, since it could do some damage if it operated at anything like 100 percent efficiency.

Improving the efficiency at least to one or two percent is quite possible, but will absorb quite a lot of your time.

To do this, you will probably want to buy a spare SCR and test the one you have to destruction with a shorter coil using thicker wire. You may also want to experiment with bigger and better capacitors, or a higher voltage circuit.

You can find much higher power SCRs, but they will be expensive, unless you get them from a surplus store. Certain online stores sell surplus electronics, and eBay usually has a selection of high-power SCRs

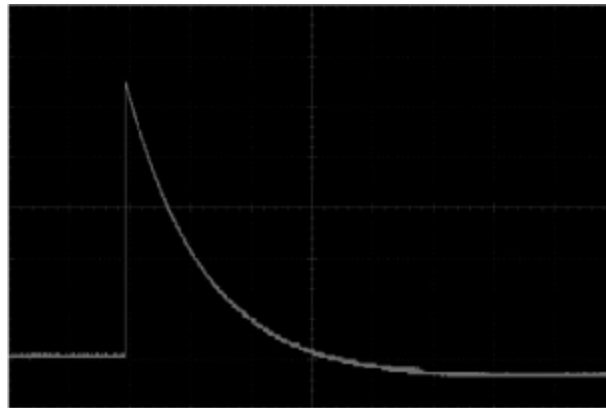


Figure 1-24 Oscilloscope trace of a test firing

You may find that taking apart the soft iron laminations from a transformer and layering them around the coil will increase the efficiency considerably.

Whatever techniques you decide to employ, be careful, and expect to destroy the occasional SCR. As well as being cautious, you need to be scientific in your approach to improving the performance of the coil gun. Keep a log book and take speed measurements after each change you make. Don't forget that if you change the coil, or the capacitors, you may find that the best starting position for the projectile may have changed. You will probably have to do tests at several different positions after each change.